Human-AI Interaction:

Multimodal Information Processing

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Overview

This lecture covers the **computational** side of multimodal AI:

1. From human perception to machine processing
2. Three foundational principles & six core challenges
3. Representation, alignment, reasoning, generation, transference, quantification
4. Designing multisensory experiences (Obrist & Velasco, 2025)
5. Fusion methods and architectures in practice
6. Benchmarking, evaluation, and open questions

`Liang, Zadeh & Morency (2024), Foundations & Trends in MML*,` *`ACM Computing Su`*

What is a Modality?

A **modality** is a way in which a natural phenomenon is perceived or expressed.

Modality	Sensor	Example
Speech / Audio	Microphone	Spoken words, tone of voice
Image / Video	Camera	Facial expressions, gestures
Text	Keyboard	Transcripts, captions
Touch / Haptics	Force sensor	Grip pressure, surface texture
Physiological	EEG, ECG	Heart rate, brain activity

Why Multimodal?

Humans naturally integrate multiple senses. Machines benefit too:

- **Affective computing**: sentiment from speech + face + words
- **Healthcare**: diagnosis from imaging + lab tests + clinical notes
- **Robotics**: manipulation with vision + touch
- **HCI**: UI understanding from screenshots + layout structure
- **Multimedia**: movie genre from video + audio + subtitles

> “ Single modalities are often noisy, ambiguous, or incomplete — combining them produces more **robust** and **accurate** systems. ”

From Human Perception to Machine Processing

The previous lecture (*Multisensory Communicative AI*) introduced the **human** side:

- Sensation → Perception → Integration
- Temporal, Spatial, and Inverse Effectiveness principles
- Illusions as evidence of integration (McGurk, Ventriloquist, Rubber Hand)

This lecture asks: **how do machines replicate — and extend — this capability?**

Bridging Human Principles & Computation

Human Principle	Machine Challenge
Temporal principle — stimuli integrated when co-occurring	Continuous **alignment** (dynamic time warping, segmentation)
Spatial principle — stimuli integrated when co-located	Discrete **alignment** (grounding, region-word matching)
Inverse effectiveness — weak unimodal → strong integration	**Robustness** and **transference** to low-resource modalities
McGurk effect — AV synergy creates new percept	Modality **interactions** (synergy, emergence)
Sensory dominance — one sense overrides others	Modality **bias** and quantification of task relevance

Three Foundational Principles

(Liang et al., 2024)

Every multimodal problem is shaped by three properties of the data:

Principle	Question
Heterogeneity	How different are the modalities from each other?
Connections	How are elements across modalities related?
Interactions	How do modalities *jointly* influence the task?

These principles motivate all six core technical challenges.

Principle 1 — Modality Heterogeneity

Modalities differ across six dimensions:

Dimension	Description	Example
Element	Basic unit of the modality	Pixel vs. word vs. Hz sample
Distribution	Frequency of elements	Words follow Zipf's law
Structure	How elements compose	Spatial (image) vs. sequential (audio)
Information	Total content per modality	Dense video vs. sparse label
Noise	Type of corruption	Occlusion (vision) vs. background noise (audio)
Task relevance	How useful the modality is	Sarcasm: tone > literal words

> “ Heterogeneity makes multimodal processing fundamentally harder than unimodal processing. ”

Principle 2 — Modality Connections

How are elements in different modalities related?

- **Statistical association**: co-occurrence (lip movement ↔ speech sound)
- **Statistical dependence**: causal or confounding relationship (emotion → voice pitch AND facial expression)
- **Semantic correspondence**: same concept, different surface form ("dog" ↔ image of dog)
- **Semantic relations**: broader, relational meaning ("red" ↔ visual redness, danger, emotion)

> Connections are the **substrate** on which alignment and reasoning operate.

Principle 3 — Modality Interactions

How do modalities jointly affect the prediction?

$$I(M_1; M_2; Y) = \underbrace{R}_{\text{redundancy}} + \underbrace{U_1 + U_2}_{\text{uniqueness}} + \underbrace{S}_{\text{synergy}}$$

Type	Meaning	Example
Redundancy	Both modalities carry same info	Face + voice both signal sadness
Uniqueness	One modality carries info the other lacks	Sarcasm: tone says "no", words say "yes"
Synergy	New info only emerges *together*	McGurk effect: AV beat audio or video alone

Interactions: Concrete Example

Task: predict sentiment from a video clip

```
"This is great"   (words = positive)
[flat tone]       (audio = neutral)
[frown]           (face = negative)
```

- If words dominate → **false positive**
- Integrating all three → recognises **sarcasm**

This is a case of **unique** information in the non-verbal channels and **synergistic** interaction across all three modalities.

Six Core Technical Challenges

(Liang et al., 2024 — taxonomy)

#	Challenge	Core Question
1	**Representation**	How do we encode and combine multimodal data?
2	**Alignment**	How do we link elements across modalities?
3	**Reasoning**	How do we compose multimodal knowledge?
4	**Generation**	How do we produce coherent multimodal output?
5	**Transference**	How do we transfer knowledge across modalities?
6	**Quantification**	How do we measure heterogeneity and interactions?

Challenge 1 — Representation

Three sub-challenges:

Fusion — integrate two or more modalities into a joint representation

$$z_{mm} = f(x_1, x_2, \ldots, x_M)$$

Coordination — keep modalities separate but align them in a shared space

$$\mathrm{sim}(\phi_1(x_1), \phi_2(x_2)) \uparrow \ \text{for matched pairs}$$

Fission — decompose into disjoint factors (modality-specific + shared)

$$z = \left[z_{\text{shared}},\ z_{\text{modal-1}},\ z_{\text{modal-2}}\right]$$

Representation Fusion: Additive & Multiplicative

Additive (late/ensemble fusion):

$$z_{mm} = w_0 + w_1 x_1 + w_2 x_2$$

Multiplicative Interactions (MI):

$$z_{mm} = w_0 + w_1 x_1 + w_2 x_2 + w_3 (x_1 \times x_2)$$

- Additive = first-order polynomial; MI = second-order polynomial.
- Cross-term $w_3 (x_1 \times x_2)$ captures **moderation**: modality 1 affects how modality 2 relates to the label.

Repr. Coordination: Contrastive Learning

Goal: matched pairs → close in embedding space; unmatched pairs → far.

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{e^{\text{sim}(z_1^+, z_2^+)/\tau}}{\sum_j e^{\text{sim}(z_1, z_{2,j})/\tau}}$$

Examples: **CLIP** (image ↔ text), **wav2BerT** (audio ↔ text).

- Contrastive learning provably captures **redundant** information across views, but **not** unique or synergistic information — a known limitation.

Representation Fission — Disentanglement

Goal: separate modality-specific from shared information.

```
Input:  [audio, text, video]
Output: z_shared   ← emotion signal present in all
        z_audio    ← speaker-specific prosody
        z_text     ← linguistic content
        z_video    ← facial muscle movements
```

> Fission enables **interpretability** and **fine-grained control** (e.g., swap only the style of a modality).

Challenge 2 — Alignment

How do we find which elements across modalities correspond?

Type	Description	Method
Discrete	Given word tokens and image regions, find matches	Attention, optimal transport
Continuous	Align continuous signals (speech waveform ↔ motion capture) without segmentation boundaries	Dynamic time warping, clustering
Contextualized	Learn representations that incorporate cross-modal context	Multimodal Transformer (MULT, CLIP, Flamingo)

Alignment difficulty: long-range dependencies, ambiguous segmentation, many-to-many mappings.

Challenge 3 — Reasoning

Composing multimodal knowledge through multiple inference steps.

- **Structure modeling**: define the composition graph (hierarchical, temporal, interactive, discovered)
- **Intermediate concepts**: attention maps, discrete symbols, natural language as reasoning medium
- **Inference paradigm**: neural (soft), logical (rule-based), causal (counterfactual)
- **External knowledge**: knowledge graphs, commonsense databases

Example: Visual Question Answering requires grounding the question in the image and reasoning over object relations.

Challenge 4 — Generation

Producing multimodal output that is coherent and consistent.

Sub-task	Input → Output	Example
Summarization	Video → text	News video summary
Translation	Text → image	DALL-E, Stable Diffusion
Creation	Text prompt → video + audio	Synchronized multimedia generation

> Key challenge: **evaluation** — how do we measure cross-modal coherence? Ethical risks: deepfakes, bias amplification.

Challenge 5 — Transference

Transfer knowledge from a *resource-rich* modality to a *resource-poor* one.

- **Cross-modal transfer**: pretrain on large image-text corpus → fine-tune for touch/radar
- **Multimodal co-learning**: share representations; use pseudo-labels from one modality to supervise another
- **Model induction**: distil behaviour from a pretrained unimodal model into a new model for a different modality

Motivation: text and images have huge pretraining corpora; audio, haptics, physiological signals do not.

Challenge 6 — Quantification

Empirically and theoretically understanding multimodal learning.

- **Heterogeneity metrics**: measure how different modalities are (information content, noise topology)
- **Bias quantification**: detect and mitigate modality-specific social biases
- **Interaction quantification**: redundancy, uniqueness, synergy (Partial Information Decomposition)
- **Learning process**: why does joint multimodal training sometimes underperform unimodal? (lazy modality, gradient conflicts)
- **Robustness metrics**: *relative robustness* and *effective robustness* under noise

Designing Multisensory Experiences

(Obrist & Velasco, 2025 – Communications of the ACM)

Moving from the **ML taxonomy** to the **design perspective**:

"We call this fusion of carefully crafted sensory elements within specific events to form a desired impression a **multisensory experience**."

- Bridges **science** (multisensory perception research) and **technology** (HCI, XR, AI)
- Relevant for: affective computing, embodied AI, human-centered multimodal systems
- Raises **design and ethical** questions that complement the computational challenges

What Makes a Multisensory Experience?

Definition (Obrist & Velasco, 2025):
A structured fusion of sensory elements across events, shaped for a specific receiver to produce a desired impression.

Four conceptual components:

Component	Description
Impression	The desired overall effect or meaning to be conveyed
Event	A specific episode or moment within the experience journey
Sensory elements	The individual stimuli (visual, auditory, tactile, olfactory, gustatory)
Receiver	The person experiencing — shaped by context, background, and state

The Experience Journey

A multisensory experience is not a single event — it is a **journey**:

```
Pre-encounter  →  Primary event   →  Post-encounter
(expectation)      (sensation  +       (memory +
                    perception +        reflection)
                    cognition  +
                    emotion)
```

- **Bottom-up** processing: sensory properties attract attention
- **Top-down** processing: goals and expectations shape perception
- Individual events can be further decomposed via **micro-phenomenology**

A coffee brand experience starts before you open the package and continues long after the last sip.

Sensory Congruence and Key Concepts

When designing multisensory experiences, four perceptual concepts guide decisions:

Concept	Meaning	Example
Sensory congruence	Stimuli feel consistent across modalities	Dubbed film feels "off" — audio ≠ lip movement
Crossmodal correspondences	Systematic associations between features across senses	Round shapes ↔ sweet taste; angular shapes ↔ bitter/sour
Sensory dominance	One sense overrides others at specific moments	Vision dominates spatial judgements (ventriloquist effect)
Sensory overload	Too much stimulation overwhelms processing	Loud music + strong smell + flashing lights

The Multisensory Ecosystem

Sensory elements do not simply **add up** — they form an **interconnected network**:

$$\text{Impression} \neq \sum_i \text{SensoryElement}_i$$

- The taste of food is altered by its **colour** (crossmodal correspondence)
- A sour taste synchronized with a new character on screen creates **surprise** (temporal + semantic congruence)
- Touch and smell in a VR film create **presence** that audio-visual alone cannot

Analogous to **synergy** in ML interaction theory: new information only emerges from the combination.

The Receiver — Individual Differences

A multisensory experience is **always relative to its receiver**:

- **Sociocultural background**: colour meanings differ across cultures
- **Previous experience**: prior encounters shape expectations and appraisal
- **Personality and sensitivity**: sensory processing sensitivity, supertasters
- **Neurodiversity and disability**: sensory impairments change modality access

Machine learning systems must similarly account for **user heterogeneity** — personalization, fairness, and accessibility are not optional extras.

Where the Senses Meet Technology

Technology enables multisensory experiences along a **reality–virtuality continuum**:

Mode	Description	Example
Physical	Real sensory stimuli in real environments	Live concert, food tasting
Augmented Reality	Digital overlays on physical world	AR flowers with information
Mixed Reality	Physical objects + digital environment	Eating real food in VR colour room
Virtual Reality	Fully digital, with haptic/olfactory devices	Dark matter dome experience

Example 1 — Multisensory Eating in VR

(Obrist & Velasco, 2025)

Impression: alter taste perception without changing the food itself
Event: participants eat real food while wearing a VR headset
Sensory elements: real food (taste + smell) + VR ambient colour + VR food shape
Receiver: adults, no sensory impairments, UK residents

Crossmodal correspondences used:

- Red / pink → perceived as sweeter
- Round shapes → sweet; angular shapes → bitter/sour

Computational parallel: exploiting learned **crossmodal associations** (semantic correspondence) to influence perception — analogous to cross-modal transfer in ML.

Example 2 — Dark Matter Experience

(Obrist & Velasco, 2025)

Impression: make an *invisible*, *abstract* concept perceivable and emotionally engaging
Event: science museum dome installation, Great Exhibition Road Festival 2019
Sensory elements: scent + spatial audio + cosmic visuals + haptic vibrations — synchronized by a central computer
Receiver: museum visitors of varied scientific backgrounds

Key insight: sensory substitution can make the **imperceptible** perceivable.

Responsibilities — Three Laws

Obrist & Velasco (2025) propose **three laws for multisensory experiences**, inspired by Asimov's robotics laws:

1. **Non-harm**: A multisensory experience must not harm the receiver, physically or psychologically.
2. **Fairness**: It must treat all receivers equitably, regardless of background, ability, or identity.
3. **Transparency**: It must be transparent about its creators, its sensory elements, and its intentions.

These map directly to AI ethics: **safety, fairness, and explainability** — now extended to the sensory domain. The **AREA framework** (Anticipate, Reflect, Engage, Act) is proposed for responsible innovation in multisensory design.

Inclusive Multisensory Design

Sensory-substitution devices enable people with sensory impairments to access multisensory experiences:

- "Seeing through sound" — visual information encoded as auditory signals
- Tactile storytelling ("Touch the Story") — narrative engagement through haptic feedback

Inclusive design considerations:

- Design *for diversity* from the start, not as an afterthought
- Richer experiences for *all* when designed with diverse receivers in mind
- Addresses digital divide and equitable access to multisensory technologies

Connection to ML: fairness and robustness across user subgroups require the same inclusive mindset — consider representation in training data, not just architecture.

MultiBench: A Unified Benchmark

MultiBench (Liang et al., NeurIPS 2021) standardises multimodal research. Three evaluation axes:

1. **Performance** — accuracy, F1, AUPRC
2. **Complexity** — parameters, training time, peak memory
3. **Robustness** — performance under missing or noisy modalities

Scope	#
Datasets	15
Modalities	10
Prediction tasks	20
Research areas	6

MultiBench Datasets

Domain	Dataset	Modalities	Task
Affective	CMU-MOSI/MOSEI	Text + Audio + Video	Sentiment
Affective	MUStARD	Text + Audio + Video	Sarcasm
Healthcare	MIMIC	Static + Time-series	Mortality / ICD9
Robotics	Vision & Touch	Image + Force	Contact detection
Robotics	MuJoCo Push	Image + Proprioception	Object pose
Finance	Stocks	Multiple stock series	Price prediction
HCI	ENRICO	Screenshot + Layout	UI category
Multimedia	AV-MNIST	Image + Audio	Digit recognition
Multimedia	MM-IMDb	Image + Text	Genre classification

MMDL: The Unified Architecture

MultiBench provides one composable architecture:

```
class MMDL(nn.Module):
    def __init__(self, encoders, fusion, head):
        self.encoders = nn.ModuleList(encoders)  # one per modality
        self.fuse = fusion                        # fusion module
        self.head = head                          # task head

    def forward(self, inputs):
        reps = [enc(x) for enc, x in zip(self.encoders, inputs)]
        fused = self.fuse(reps)
        return self.head(fused)
```

Mix and match: swap any encoder, any fusion module, any task head.

Fusion Methods: Early & Late

Early Fusion: concatenate raw or feature *before* learning:

$$z = W \cdot [x_1 \,\|\, x_2 \,\|\, \cdots \,\|\, x_M] + b$$

Simple, captures low-level interactions, one modality can dominate.

Late Fusion: train unimodal predictors, aggregate predictions:

$$\hat{y} = \sum_m w_m f_m(x_m)$$

Each modality trains at its own pace, no interactions

Tensor Fusion

Tensor Fusion (Zadeh et al., 2017) captures all higher-order interactions via outer product:

$$Z_{TF} = (x_1 \oplus 1) \otimes (x_2 \oplus 1) \otimes (x_3 \oplus 1)$$

Appending 1 ensures lower-order terms are included.

For 3 modalities of size d_1, d_2, d_3:

$$Z_{TF} \in \mathbb{R}^{(d_1+1)(d_2+1)(d_3+1)}$$

Expressive but **expensive** — dimensionality grows multiplicatively.

Low-Rank Tensor Fusion

Low-Rank Tensor Fusion (2018) factorises the tensor weight:

$$W = \sum_{r=1}^{R} w_1^{(r)} \otimes w_2^{(r)} \otimes w_3^{(r)}$$

Rank $R \ll d_1 d_2 d_3 \rightarrow$ drastically fewer parameters.

```
class LowRankTensorFusion(nn.Module):
    # Projects each modality into rank-R factors
    # then combines via element-wise product and sum
```

Same expressivity target as full tensor fusion, but **tractable** at scale.

Multimodal Transformer (MulT)

MULT uses directional cross-modal attention:

For source modality β and target modality α:

$$\text{CM-Attn}_{\beta\to\alpha} : Q = x_\alpha,\ K = V = x_\beta$$

$$z_{\alpha\leftarrow\beta} = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Each token in α attends to all tokens in β at every time step — captures **long-range cross-modal dependencies** without explicit alignment.

MultiBench Evaluation

- **Complexity**: Peak memory, number of parameters, training and inference time.
- **Robustness**

$$\text{Relative Robustness} = \frac{\text{perf under noise}}{\text{perf without noise}}$$

$$\text{Effective Robustness} = \text{perf under noise} - \text{baseline (unimo}$$

Plotting both reveals the **accuracy–robustness tradeoff**: some fusion methods are accurate but brittle; others are robust but weaker.

Open Questions (Liang et al., 2024>9.1)

- **Representation**: How do we formally quantify heterogeneity and interactions? What theoretical guarantees can we give for fusion methods?
- **Alignment**: Compositionality — can models align *novel* combinations of elements?
- **Generation**: Synchronized audio-video-text creation remains unsolved; ethical risks (deepfakes, bias) need frameworks.
- **Transference**: High-modality learning (>5 modalities) — how to handle non-parallel data?
- **Quantification**: Explainability for non-visual modalities (audio, haptics, physiological) — how to interpret learned representations?

Summary

From Perception to Processing: Human principles (temporal, spatial, inverse effectiveness) ↔ alignment, robustness, transference in ML

Three Foundational Principles (Liang et al., 2024): Heterogeneity · Connections · Interactions

Six Core Challenges: Representation → Alignment → Reasoning → Generation → Transference → Quantification

Designing Multisensory Experiences (Obrist & Velasco, 2025):

- Four components: Impression · Event · Sensory elements · Receiver
- Senses meet technology on the reality–virtuality continuum
- Responsibilities: non-harm, fairness, transparency (Three Laws)

Practice (MultiBench / MultiZoo):

- MMDL: Encoders → Fusion → Head
- Fusion spectrum:
  - Early & Late Fusion
  - Low-Rank Tensor Fusion
  - MI
  - Cross-modal Attention
- Evaluation: Performance, Complexity, Robustness